

Differentiation Basic Problem

1. $y = \sqrt{x} \sin x - 6$

$$\begin{aligned}\frac{dy}{dx} &= \sqrt{x} \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(\sqrt{x}) - 0 \\ &= \sqrt{x} \cos x + \sin x \times \frac{1}{2\sqrt{x}} \text{ (Ans.)}\end{aligned}$$

2. $y = \log x + \sec x$

$$\frac{dy}{dx} = \frac{1}{x} + \sec x \tan x \text{ (Ans.)}$$

3. $y = \sqrt{x} + \sqrt[4]{x} + \frac{2}{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{x}} + \frac{1}{4}x^{1/4-1} + 2 \times \left(-\frac{1}{x^2}\right) \\ &= \frac{1}{2\sqrt{x}} + \frac{1}{4}x^{-3/4} - \frac{2}{x^2} \\ &= \frac{1}{2\sqrt{x}} + \frac{1}{4x^{3/4}} - \frac{2}{x^2} \text{ (Ans.)}\end{aligned}$$

4. $y = 7\log x - 5\log x + 8\cos x$

$$\frac{dy}{dx} = \frac{7}{x} - 5 \times \frac{1}{x} - 8 \sin x \text{ (Ans.)}$$

5. $y = (x+1)^2 (x+1)$

$$y = (x^2 + 2x + 1)(x + 1)$$

$$y = x^3 + x^2 + 2x^2 + 2x + x + 1$$

$$y = x^3 + 3x^2 + 3x + 1$$

$$\frac{dy}{dx} = 3x^2 + 6x + 3 + 0 \text{ (Ans.)}$$

6. $y = e^x \cos x + 6$

$$\begin{aligned}\frac{dy}{dx} &= e^x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(e^x) + 0 \\ &= -e^x \sin x + e^x \cos x \\ &= e^x (\cos x - \sin x) \text{ (Ans.)}\end{aligned}$$

7. $y = 2x^6 - 5e^x + 7\tan x$

$$\frac{dy}{dx} = 12x^5 - 5e^x + 7\sec^2 x \text{ (Ans.)}$$

8. $y = 8e^x \tan x$

$$\begin{aligned}\frac{dy}{dx} &= 8 \left(e^x \frac{d}{dx}(\tan x) + \tan x \frac{d}{dx}(e^x) \right) \\ &= 8 \{ e^x \sec^2 x + \tan x e^x \} \\ &= 8e^x \{ \sec^2 x + \tan x \} \text{ (Ans.)}\end{aligned}$$

9. $y = \log x \cdot \cos x$

$$\begin{aligned}\frac{dy}{dx} &= \log x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(\log x) \\ &= \log x (-\sin x) + \cos x \times \frac{1}{x} \\ &= -\sin x \log x + \frac{\cos x}{x} \text{ (Ans.)}\end{aligned}$$

10. $y = \tan x \sec x$

$$\begin{aligned}\frac{dy}{dx} &= \sec^2 x \cdot \sec x + \tan x \cdot \sec x \cdot \tan x \\ &= \sec^3 x + \tan^2 x \sec x \\ &= \sec x (\sec^2 x + \tan^2 x) \text{ (Ans.)}\end{aligned}$$

11. $y = 4\sqrt{x} \sin^{-1} x$

$$\begin{aligned}\frac{dy}{dx} &= 4 \left\{ \sqrt{x} \times \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x \cdot \frac{1}{2\sqrt{x}} \right\} \\ &= 4 \left\{ \frac{\sqrt{x}}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{2\sqrt{x}} \right\} \text{ (Ans.)}\end{aligned}$$

12. $y = \frac{\ln x}{\cos x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos x \frac{d}{dx}(\ln x) - \ln x \times \frac{d}{dx}(\cos x)}{\cos^2 x} \\ &= \frac{\cos x \times \frac{1}{x} - \ln x \times (-\sin x)}{\cos^2 x} \\ &= \frac{\frac{\cos x}{x} + \sin x \ln x}{\cos^2 x} \text{ (Ans.)}\end{aligned}$$

13. $y = \frac{1+\sin x}{1+\cos x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1+\cos x) \cos x - (1+\sin x) \times (-\sin x)}{(1+\cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin x + \sin^2 x}{(1+\cos x)^2} \text{ (Ans.)}\end{aligned}$$

14. $y = \frac{\sin x}{x+\cos x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x+\cos x) \cos x - \sin x (1-\sin x)}{(x+\cos x)^2} \\ &= \frac{(x+\cos x) \cos x - \sin x + \sin^2 x}{(x+\cos x)^2} \\ &= \frac{x \cos x + \cos^2 x - \sin x + \sin^2 x}{(x+\cos x)^2} \\ &= \frac{x \cos x - \sin x + 1}{(x+\cos x)^2} \text{ (Ans.)}\end{aligned}$$

$$\begin{aligned}
 15. y &= \frac{\sqrt{x} + 1}{\sqrt{x} - 1} \\
 \frac{dy}{dx} &= \frac{(\sqrt{x}-1) \cdot \frac{1}{2\sqrt{x}} - (\sqrt{x}+1) \times \frac{1}{2\sqrt{x}}}{(\sqrt{x}-1)^2} \\
 &= \frac{\frac{1}{2\sqrt{x}}(\sqrt{x}-1-\sqrt{x}-1)}{(\sqrt{x}-1)^2} \\
 &= \frac{\frac{1}{2\sqrt{x}} \times (-2)}{(\sqrt{x}-1)^2} \\
 &= \frac{-\frac{1}{\sqrt{x}}}{(\sqrt{x}-1)^2} \\
 &= -\frac{1}{\sqrt{x}(\sqrt{x}-1)^2} \text{ (Ans.)}
 \end{aligned}$$

Home Work:

1. $y = \frac{x}{\log x}$

2. $y = \frac{x-1}{x+1}$

3. $y = \frac{5x+3}{2x-1}$

4. $y = \frac{2x}{x+1}$

5. $y = \frac{2x+1}{3x+1}$

6. $y = \frac{x+1}{x+2}$

7. $y = \frac{1-\cos x}{1-\sin x}$

8. $y = \frac{\sec x}{8 \tan x}$

Chain Rule

1. $y = \sin 5x$

$$\begin{aligned}
 \frac{dy}{dx} &= \cos 5x \cdot \frac{d}{dx}(5x) \\
 &= 5 \cos 5x \text{ (Ans.)}
 \end{aligned}$$

2. $y = \log \sin x$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{\sin x} \times \frac{d}{dx}(\sin x) \\
 &= \frac{1}{\sin x} \times \cos x \\
 &= \cot x \text{ (Ans.)}
 \end{aligned}$$

3. $y = 8 \cos \sqrt{x}$

$$\begin{aligned}
 \frac{dy}{dx} &= 8 (-\sin \sqrt{x}) \cdot \frac{1}{2\sqrt{x}} \\
 &= \frac{-8 \sin \sqrt{x}}{2\sqrt{x}} \\
 &= \frac{-4 \sin \sqrt{x}}{\sqrt{x}} \text{ (Ans.)}
 \end{aligned}$$

4. $y = \sec(\sec x)$

$$\begin{aligned}
 \frac{dy}{dx} &= \sec(\sec x) \cdot \tan(\sec x) \cdot \frac{d}{dx}(\sec x) \\
 &= \sec(\sec x) \cdot \tan(\sec x) \cdot \sec x \tan x \text{ (Ans.)}
 \end{aligned}$$

5. $y = (\sin x)^2$

$$\begin{aligned}
 \frac{dy}{dx} &= 2 \sin x \cdot \frac{d}{dx}(\sin x) \\
 &= 2 \sin x \cos x \\
 &= \sin 2x \text{ (Ans.)}
 \end{aligned}$$

6. $y = \log(\sin 5x)$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{\sin 5x} \times \frac{d}{dx}(\sin 5x) \\
 &= \frac{1}{\sin 5x} \times \cos 5x \cdot \frac{d}{dx}(5x) \\
 &= \cot 5x \cdot (5 \cdot 1) \\
 &= 5 \cot 5x \text{ (Ans.)}
 \end{aligned}$$

7. $y = \tan(\cos x)$

$$\begin{aligned}
 \frac{dy}{dx} &= \sec^2(\cos x) \times \frac{d}{dx}(\cos x) \\
 &= \sec^2(\cos x) \times (-\sin x) \\
 &= -\sin x \sec^2(\cos x) \text{ (Ans.)}
 \end{aligned}$$

8. $y = \log \sec(5x)$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{\sec 5x} \cdot \frac{d}{dx}(\sec 5x) \\
 &= \frac{1}{\sec 5x} \times \sec 5x \cdot \tan 5x \cdot 5 \\
 &= 5 \tan 5x \text{ (Ans.)}
 \end{aligned}$$

9. $y = \sec(5x+3)$

$$\begin{aligned}
 \frac{dy}{dx} &= \sec(5x+3) \cdot \tan(5x+3) \cdot \frac{d}{dx}(5x+3) \\
 &= 5 \sec(5x+3) \tan(5x+3) \text{ (Ans.)}
 \end{aligned}$$

$$10. y = \sqrt{\sin x}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{\sin x}} \times \frac{d}{dx} (\sin x) \\ &= \frac{1}{2\sqrt{\sin x}} \times \cos x \quad (\text{Ans.})\end{aligned}$$

$$11. y = \cos (\log 5x)$$

$$\begin{aligned}\frac{dy}{dx} &= -\sin (\log 5x) \cdot \frac{d}{dx} (\log 5x) \\ &= -\sin (\log 5x) \times \frac{1}{5x} \times \frac{d}{dx} 5x \\ &= -5 \sin (\log 5x) \times \frac{1}{5x} \\ &= -\frac{1}{x} \times \sin (\log 5x) \quad (\text{Ans.})\end{aligned}$$

$$12. y = e^{\sqrt{\sin x}}$$

$$\begin{aligned}\frac{dy}{dx} &= e^{\sqrt{\sin x}} \cdot \frac{d}{dx} (\sqrt{\sin x}) \\ &= e^{\sqrt{\sin x}} \times \frac{1}{2\sqrt{\sin x}} \cos x \quad (\text{Ans.})\end{aligned}$$

$$13. y = \tan (\cos 6x)$$

$$\begin{aligned}\frac{dy}{dx} &= \sec^2 (\cos 6x) \cdot \frac{d}{dx} (\cos 6x) \\ &= \sec^2 (\cos 6x) \times (-\sin 6x) \times \frac{d}{dx} (6x) \\ &= -6 \sec^2 (\cos 6x) \times \sin 6x \quad (\text{Ans.})\end{aligned}$$

$$14. y = \sqrt{e^{\sqrt{x}}}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times \frac{d}{dx} e^{\sqrt{x}} \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times e^{\sqrt{x}} \cdot \frac{d}{dx} (\sqrt{x}) \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times e^{\sqrt{x}} \times \frac{1}{2\sqrt{x}} \\ &= \frac{e^{\sqrt{x}}}{4\sqrt{x} \sqrt{e^{\sqrt{x}}}} \quad (\text{Ans.})\end{aligned}$$

$$15. y = \cot (\log \sqrt[3]{x})$$

$$\begin{aligned}\frac{dy}{dx} &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \cdot \frac{d}{dx} (\log \sqrt[3]{x}) \\ &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \cdot \frac{d}{dx} (\sqrt[3]{x}) \\ &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \times \frac{1}{3} x^{1/3-1} \\ &= -\frac{1}{3} \operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \times x^{-2/3} \\ &= -\frac{1}{3 \sqrt[3]{x} \cdot x^{2/3}} \times \operatorname{cosec}^2 (\log \sqrt[3]{x}) \\ &= -\frac{1}{3x} \times \operatorname{cosec}^2 (\log \sqrt[3]{x}) \quad (\text{Ans.})\end{aligned}$$

$$16. y = (4x + 3)^2$$

$$\begin{aligned}\frac{dy}{dx} &= 2(4x+3) \cdot 4 \\ &= 8(4x+3) \quad (\text{Ans.})\end{aligned}$$

$$17. y = \frac{2}{(x^4 + 2)^3}$$

$$y = 2 \times (x^4 + 2)^{-3}$$

$$\begin{aligned}\frac{dy}{dx} &= 2(-3)(x^4 + 2)^{-4} \times \frac{d}{dx} (x^4 + 2) \\ &= -6(x^4 + 2)^{-4} \cdot 4x^3\end{aligned}$$

$$18. y = \sin \frac{e^x}{x^2}$$

$$\begin{aligned}\frac{dy}{dx} &= \cos \left(\frac{e^x}{x^2} \right) \cdot \frac{d}{dx} \left(\frac{e^x}{x^2} \right) \\ &= \cos \left(\frac{e^x}{x^2} \right) \times \frac{x^2 e^x - 2x e^x}{(x^2)^2} \\ &= \cos \left(\frac{e^x}{x^2} \right) \times \frac{e^x (x^2 - 2x)}{x^4} \quad (\text{Ans.})\end{aligned}$$

$$19. y = \sin (e^x \cdot x^2)$$

$$\begin{aligned}\frac{dy}{dx} &= \cos (e^x \cdot x^2) \cdot \frac{d}{dx} (e^x \cdot x^2) \\ &= \cos (e^x \cdot x^2) \cdot (e^x 2x + x^2 e^x) \\ &= e^x (2x + x^2) \cos (e^x x^2) \quad (\text{Ans.})\end{aligned}$$

$$20. y = (x^2 + 5)^3 \cdot (x^3 - 1)^4$$

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 5)^3 \cdot \frac{d}{dx} (x^3 - 1)^4 + (x^3 - 1)^4 \cdot \frac{d}{dx} (x^2 + 5)^3 \\ &= (x^2 + 5)^3 \cdot 4(x^3 - 1)^3 \cdot 3x^2 + (x^3 - 1)^4 \cdot 3(x^2 + 5)^2 \cdot 2x \\ &= 12x^2 (x^3 - 1)^3 (x^2 + 5)^3 + 6x (x^3 - 1)^4 (x^2 + 5)^2 \quad (\text{Ans.})\end{aligned}$$

$$21. y = \sqrt[3]{\sin x - 1}$$

$$y = (\sin x - 1)^{1/3}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{3} (\sin x - 1)^{1/3-1} \times \cos x \\ &= \frac{1}{3} \cos x (\sin x)^{-2/3} \\ &= \frac{\cos x}{3 (\sin x)^{2/3}} \quad (\text{Ans.})\end{aligned}$$

$$22. y = e^{\sqrt{x}}$$

$$\frac{dy}{dx} = e^{\sqrt{x}}$$

$$= e^{\sqrt{x}}$$

$$= e^{\sqrt{x}}$$

$$1. y =$$

$$2. y =$$

$$3. y =$$

$$4. y =$$

$$5. y =$$

$$6. y =$$

$$7. y =$$

$$8. y =$$

$$9. y =$$

$$10. y =$$

$$11. y =$$

$$12. y =$$

$$13. y =$$

$$14. y =$$

$$15. y =$$

$$16. y =$$

$$10. y = \sqrt{\sin x}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{\sin x}} \times \frac{d}{dx} (\sin x) \\ &= \frac{1}{2\sqrt{\sin x}} \times \cos x \quad (\text{Ans.})\end{aligned}$$

$$11. y = \cos (\log 5x)$$

$$\begin{aligned}\frac{dy}{dx} &= -\sin (\log 5x) \frac{d}{dx} (\log 5x) \\ &= -\sin (\log 5x) \times \frac{1}{5x} \times \frac{d}{dx} 5x \\ &= -5 \sin (\log 5x) \times \frac{1}{5x} \\ &= -\frac{1}{x} \times \sin (\log 5x) \quad (\text{Ans.})\end{aligned}$$

$$12. y = e^{\sqrt{\sin x}}$$

$$\begin{aligned}\frac{dy}{dx} &= e^{\sqrt{\sin x}} \cdot \frac{d}{dx} (\sqrt{\sin x}) \\ &= e^{\sqrt{\sin x}} \times \frac{1}{2\sqrt{\sin x}} \cos x \quad (\text{Ans.})\end{aligned}$$

$$13. y = \tan (\cos 6x)$$

$$\begin{aligned}\frac{dy}{dx} &= \sec^2 (\cos 6x) \frac{d}{dx} (\cos 6x) \\ &= \sec^2 (\cos 6x) \times (-\sin 6x) \times \frac{d}{dx} (6x) \\ &= -6 \sec^2 (\cos 6x) \times \sin 6x \quad (\text{Ans.})\end{aligned}$$

$$14. y = \sqrt{e^{\sqrt{x}}}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times \frac{d}{dx} e^{\sqrt{x}} \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times e^{\sqrt{x}} \frac{d}{dx} (\sqrt{x}) \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \times e^{\sqrt{x}} \times \frac{1}{2\sqrt{x}} \\ &= \frac{e^{\sqrt{x}}}{4\sqrt{x}\sqrt{e^{\sqrt{x}}}} \quad (\text{Ans.})\end{aligned}$$

$$15. y = \cot (\log \sqrt[3]{x})$$

$$\begin{aligned}\frac{dy}{dx} &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \cdot \frac{d}{dx} (\log \sqrt[3]{x}) \\ &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \frac{d}{dx} (\sqrt[3]{x}) \\ &= -\operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \times \frac{1}{3} x^{1/3-1} \\ &= -\frac{1}{3} \operatorname{cosec}^2 (\log \sqrt[3]{x}) \times \frac{1}{\sqrt[3]{x}} \times x^{-2/3} \\ &= -\frac{1}{3 \sqrt[3]{x} x^{2/3}} \times \operatorname{cosec}^2 (\log \sqrt[3]{x}) \\ &= -\frac{1}{3x} \times \operatorname{cosec}^2 (\log \sqrt[3]{x}) \quad (\text{Ans.})\end{aligned}$$

$$16. y = (4x + 3)^2$$

$$\begin{aligned}\frac{dy}{dx} &= 2(4x+3) \cdot 4 \\ &= 8(4x+3) \quad (\text{Ans.})\end{aligned}$$

$$17. y = \frac{2}{(x^4+2)^3}$$

$$y = 2 \times (x^4 + 2)^{-3}$$

$$\begin{aligned}\frac{dy}{dx} &= 2(-3)(x^4 + 2)^{-4} \times \frac{d}{dx} (x^4 + 2) \\ &= -6(x^4 + 2)^{-4} \cdot 4x^3\end{aligned}$$

$$18. y = \sin \frac{e^x}{x^2}$$

$$\begin{aligned}\frac{dy}{dx} &= \cos \left(\frac{e^x}{x^2} \right) \frac{d}{dx} \left(\frac{e^x}{x^2} \right) \\ &= \cos \left(\frac{e^x}{x^2} \right) \times \frac{x^2 e^x - 2x e^x}{(x^2)^2} \\ &= \cos \left(\frac{e^x}{x^2} \right) \times \frac{e^x (x^2 - 2x)}{x^4} \quad (\text{Ans.})\end{aligned}$$

$$19. y = \sin (e^x \cdot x^2)$$

$$\begin{aligned}\frac{dy}{dx} &= \cos (e^x \cdot x^2) \cdot \frac{d}{dx} (e^x \cdot x^2) \\ &= \cos (e^x \cdot x^2) \cdot (e^x 2x + x^2 e^x) \\ &= e^x (2x + x^2) \cos (e^x x^2) \quad (\text{Ans.})\end{aligned}$$

$$20. y = (x^2 + 5)^3 \cdot (x^3 - 1)^4$$

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 5)^3 \frac{d}{dx} (x^3 - 1)^4 + (x^3 - 1)^4 \frac{d}{dx} (x^2 + 5)^3 \\ &= (x^2 + 5)^3 4(x^3 - 1)^3 \cdot 3x^2 + (x^3 - 1)^4 \times 3(x^2 + 5)^2 \times 2x \\ &= 12x^2 (x^3 - 1)^3 (x^2 + 5)^3 + 6x (x^3 - 1)^4 (x^2 + 5)^2 \quad (\text{Ans.})\end{aligned}$$

$$21. y = \sqrt[3]{\sin x - 1}$$

$$y = (\sin x - 1)^{1/3}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{3} (\sin x - 1)^{1/3-1} \times \cos x \\ &= \frac{1}{3} \cos x (\sin x)^{-2/3} \\ &= \frac{\cos x}{3 (\sin x)^{2/3}} \quad (\text{Ans.})\end{aligned}$$

$$\begin{aligned}
 22. y &= e^{\sqrt{x}} \cdot \sqrt{x^2 - 1} \\
 \frac{dy}{dx} &= e^{\sqrt{x}} \cdot \frac{d}{dx}(\sqrt{x^2 - 1}) + (\sqrt{x^2 - 1}) \frac{d}{dx}(e^{\sqrt{x}}) \\
 &= e^{\sqrt{x}} \times \frac{1}{2\sqrt{x^2 - 1}} \times 2x + (\sqrt{x^2 - 1}) \times e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} \\
 &= e^{\sqrt{x}} \left\{ \frac{x}{\sqrt{x^2 - 1}} + \frac{(\sqrt{x^2 - 1})}{2\sqrt{x}} \right\} \text{(Ans.)}
 \end{aligned}$$

Home Work:

1. $y = (5x - 3)^4$
2. $y = (3x^2 + 4)^5$
3. $y = \sqrt{2x^3 - 1}$
4. $y = \sin(3x + 4)$
5. $y = \cos(5x - 1)$
6. $y = \sin^3 x$
7. $y = 3\tan^2 x$
8. $y = \sin\{\cos(\log x^3)\}$
9. $y = \sin(x^3 + x^2 + 8)$
10. $y = (x^3 + 2x - 1)^5$
11. $y = (2x^{3/2} - 3x^{4/3} - 5)^{5/2}$
12. $y = \sqrt{x^2 + \sqrt{x^2 + 1}}$
13. $y = \left(\frac{x^2 + 1}{x^2 - 1}\right)^3$
14. $y = e^{2x+1} \cdot \sin^3 x$
15. $y = (1 + \cos 2x)^2$
16. $y = \sqrt{x + \frac{1}{x}}$

17. $y = (3x - 2)(5x + 4)^4$
18. $y = \frac{(4x - 3)^5}{2x - 1}$
19. $y = \sqrt{\frac{2x}{x^3 - 2x}}$
20. $y = \log\left(\frac{x+1}{x-1}\right)$
21. $y = \sqrt[3]{\ln(\sin x)}$
22. $y = \log(\log(\log x))$
23. $y = \sin^2 x^2$
24. $y = \frac{\cos^2 x}{\cos x^2}$
25. $y = \frac{\sqrt{1 - \sqrt{x}}}{\sqrt{1 + \sqrt{x}}}$
26. $y = \sqrt{e^x + 1}$
27. $y = \sin^3(2x + 3)$
28. $y = \sin^2 x + \cos^3 x$
29. $y = \log(\cos(e^{3x}))$
30. $y = \tan^4(\sec^2 x)$

Explicit and Implicit1. Find $\frac{dy}{dx}$ from given function

$$ax^2 + 2hxy + by^2 + d = 0$$

Solution: Given,

$$ax^2 + 2hxy + by^2 + d = 0$$

Diff w.r. to x

$$a.2x + 2h(x \frac{dy}{dx} + y) + b.2y \frac{dy}{dx} = 0$$

$$\text{or, } (2hx + 2by) \frac{dy}{dx} = -2ax - 2hy$$

$$\text{or, } \frac{dy}{dx} = \frac{-(ax+hy)}{hx+by} \quad (\text{Ans.})$$

2. Find $\frac{dy}{dx}$ from given function

$$y = e^{\sin x} \sin x$$

Solution:

Taking log on both sides,

$$\text{Log } y = \log(e^{\sin x} \cdot \sin x) = \sin x + \log(\sin x)$$

Diff w.r. to x

$$\frac{1}{y} \frac{dy}{dx} = \cos x + \frac{1}{\sin x} \cdot \cos x \cdot a^x \log a$$

$$\therefore \frac{dy}{dx} = y (\cos x + a^x \cot x \log a)$$

$$= e^{\sin x} \cdot \sin x (\cos x + a^x \cot x \log a) \quad (\text{Ans.})$$

3. Find $\frac{dy}{dx}$ from given function

$$y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$$

Taking log on both sides,

$$\text{Log } y = \log e^{x^2} + \log \tan^{-1} x - \log \sqrt{1+x^2}$$

$$= x^2 + \log \tan^{-1} x - \frac{1}{2} \log(1+x^2)$$

Diff w.r. to x

$$\frac{1}{y} \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x} \cdot \frac{1}{1+x^2} - \frac{1}{2} \cdot \frac{1}{1+x^2} \cdot 2x$$

$$\therefore \frac{dy}{dx} = y \left(2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2} \right)$$

$$= \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left(2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2} \right)$$

Ans

4. Find $\frac{dy}{dx}$ from given function

$$\log(xy) = x^2 + y^2$$

Given, $\log(xy) = x^2 + y^2$

$$\text{or, } \log x + \log y = x^2 + y^2$$

Diff w.r. to x

$$\frac{1}{x} + \frac{1}{y} \frac{dy}{dx} = 2x + 2y \frac{dy}{dx}$$

$$\rightarrow \left(\frac{1}{y} - 2y \right) \frac{dy}{dx} = 2x - \frac{1}{x}$$

$$\rightarrow \frac{dy}{dx} = \frac{2x - \frac{1}{x}}{\frac{1}{y} - 2y}$$

$$= \frac{y(2x^2 - 1)}{x(1 - 2y^2)} \quad (\text{Ans.})$$

5. Find $\frac{dy}{dx}$ from given function

$$y = \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$$

Given,

$$y = \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$$

$$= \tan^{-1} \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}}$$

$$= \tan^{-1} \tan \frac{x}{2}$$

$$= \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2} \quad (\text{Ans.})$$

6. Find $\frac{dy}{dx}$ from given function

$$\sin y = x \cdot \sin(a+y) \quad [\text{Oct-16}]$$

Solution:

Diff w.r. to x

$$\cos y \frac{dy}{dx} = x \cos(a+y) \frac{dy}{dx} + \sin(a+y)$$

$$\text{Or, } [\cos y - x \cos(a+y)] \frac{dy}{dx} = \sin(a+y)$$

$$\therefore \frac{dy}{dx} = \frac{\sin(a+y)}{\cos y - x \cos(a+y)} \quad (\text{Ans})$$

7. Find $\frac{dy}{dx}$ from given function

$$y = \frac{\tan x}{x} \log \frac{e^x}{x^2}$$

$$= \frac{\tan x}{x} (\log e^x - \log x^2)$$

$$= \frac{\tan x}{x} (x - 2 \log x)$$

$$= \tan x - \frac{2 (\tan x \log x)}{x}$$

$$\therefore \frac{dy}{dx} = \sec^2 x - \frac{2 \{ x \cdot \frac{d}{dx} (\tan x \log x) + \tan x \cdot \log x \}}{x^2}$$

$$= \sec^2 x - \frac{2 \log x \sec^2 x}{x} - \frac{2 \tan x}{x^2} - \frac{2 \tan x \log x}{x^2}$$

(Ans)

8. Find $\frac{dy}{dx}$ from given function [Oct-17]

$$y = \sin \left\{ 2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right\}$$

$$= \sin \left\{ 2 \tan^{-1} \sqrt{\frac{1-\cos \theta}{1+\sin \theta}} \right\}$$

$$= \sin \left\{ 2 \tan^{-1} \sqrt{\frac{2 \sin^2 \theta / 2}{2 \cos^2 \theta / 2}} \right\}$$

$$= \sin \left\{ 2 \tan^{-1} \tan \theta / 2 \right\}$$

$$Y = \sin \theta$$

$$= \sqrt{1-x^2}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} \times -2x$$

$$= \frac{-x}{\sqrt{1-x^2}} \text{ (Ans.)}$$

$$\text{let } x = \cos \theta$$

$$\sin \theta = \sqrt{1-x^2}$$

9. Find $\frac{dy}{dx}$ from given function

If $y =$

$$\sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}, \text{ then}$$

$$\text{show that } \frac{dy}{dx} = \frac{\cos x}{2y-x} \text{ [Ap-19, Ap-18]}$$

Ans.

$$y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$$

$$y = \sqrt{\sin x + y}$$

$$y^2 = \sin x + y \text{ [squing both side]}$$

Diff w.r. to x,

$$2y \cdot \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$\frac{dy}{dx} (2y-1) = \cos x$$

$$\frac{dy}{dx} = \frac{\cos x}{2y-1} \text{ [Proved]}$$

April 17:

10. Find $\frac{dy}{dx}$ at $x=1$, $y=8$ where $\sqrt[3]{x} + \sqrt[3]{y} = 3$

Ans. Given,

$$\sqrt[3]{x} + \sqrt[3]{y} = 3$$

$$x^{1/3} + y^{1/3} = 3$$

Differentiation both side w.r. to x,

$$\frac{1}{3} x^{-2/3} + \frac{1}{3} y^{-2/3} \cdot \frac{dy}{dx} = 0$$

$$\frac{1}{3} + \frac{1}{3} \cdot (8)^{-2/3} \frac{dy}{dx} = 0$$

$$\frac{1}{3} (2^3)^{-2/3} \frac{dy}{dx} = -1/3$$

$$\left(\frac{1}{3 \times 2^2} \right) \frac{dy}{dx} = -1/3$$

$$\frac{dy}{dx} = \frac{-12}{3} = -4 \text{ (Ans.)}$$

11. Find $\frac{dy}{dx}$ from given function

$$y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$$

Let,

$$y = u + v,$$

$$\text{where } u = (\tan x)^{\cot x}, v = (\cot x)^{\tan x}$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \dots \dots \dots (i)$$

$$\text{Now, } u = (\tan x)^{\cot x}$$

$$\rightarrow \log u = \cot x \log (\tan x)$$

Diff w.r. to x,

$$\frac{1}{u} \frac{du}{dx} = \cot x \cdot \frac{1}{\tan x} \cdot \sec^2 x + \log (\tan x) (-\operatorname{cosec}^2 x)$$

$$\frac{du}{dx} = u [\sec^2 x \cdot \cot^2 x - \operatorname{cosec}^2 x \log (\tan x)]$$

$$\text{Again, } v = (\cot x)^{\tan x}$$

$$\log v = \tan x \log (\cot x)$$

$$\rightarrow \frac{1}{v} \frac{dv}{dx} = \tan x \cdot \frac{1}{\cot x} \cdot (-\operatorname{cosec}^2 x) + \log (\cot x)$$

$$\sec^2 x$$

$$\rightarrow \frac{dv}{dx} = v [-\operatorname{cosec}^2 x \tan^2 x + \sec^2 x \log (\cot x)]$$

Substituting the value of $\frac{du}{dx}$ and $\frac{dv}{dx}$ on Eq(i)

$$\frac{dy}{dx} = (\tan x)^{\cot x} [\sec^2 x \cdot \cot^2 x - \operatorname{cosec}^2 x \log (\tan x)]$$

$$+ (\cot x)^{\tan x} [-\operatorname{cosec}^2 x \tan^2 x + \sec^2 x \log (\cot x)]$$

(Ans.)

12. Find $\frac{dy}{dx}$ from given function

$$y = \sin(x^{\log x}) + \sin^2(\cos^{-1}x)$$

$$\begin{aligned}\text{Given, } y &= \sin(x^{\log x}) + \sin^2(\cos^{-1}x) \\ &= \sin(x^{\log x}) + 1 - \cos^2(\cos^{-1}x) \\ &= \sin(x^{\log x}) + 1 - x^2\end{aligned}$$

Diff w.r. to x

$$\frac{dy}{dx} = \cos(x^{\log x}) \frac{d}{dx} x^{\log x} - 2x \dots \dots \dots (i)$$

$$\text{Let, } u = x^{\log x}$$

$$\rightarrow \log u = \log x \cdot \log x$$

$$\rightarrow \log u = (\log x)^2$$

Diff w.r. to x

$$\begin{aligned}\frac{1}{u} \frac{du}{dx} &= 2 \log x \cdot \frac{1}{x} \\ \frac{du}{dx} &= x^{\log x} \cdot \frac{2 \log x}{x}\end{aligned}$$

$$\text{From (i), } \frac{dy}{dx} = \cos(x^{\log x}) \cdot x^{\log x} \cdot \frac{2 \log x}{x} - 2x \text{ (Ans.)}$$

13. Find $\frac{dy}{dx}$ from given function

$$y = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$$

Sol:

$$\text{Let, } x = \tan \theta$$

Given,

$$y = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$$

$$Y = \tan^{-1} \frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta}$$

$$= \tan^{-1} \frac{\sec \theta - 1}{\tan \theta}$$

$$= \tan^{-1} \frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}}$$

$$= \tan^{-1} \frac{1 - \cos \theta}{\sin \theta}$$

$$= \tan^{-1} \frac{2 \sin^2 \theta / 2}{2 \sin \theta / 2 \cos \theta / 2}$$

$$= \tan^{-1} \tan \theta / 2$$

$$= \theta / 2$$

$$= \frac{1}{2} \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{1}{(1+x^2)} \text{ (Ans.)}$$

14. Find $\frac{dy}{dx}$ from given function

$$(\cos x)^y + (\sin y)^x = 0$$

Ans:

$$\text{Let, } u = (\cos x)^y$$

$$v = (\sin y)^x$$

$$\therefore u + v =$$

$$\frac{du}{dx} + \frac{dv}{dx} = 0 \dots \dots \dots (1)$$

$$\text{Now, } u = (\cos x)^y$$

taking log on both sides,

$$\log u = y \log \cos x$$

Diff w.r. to x,

$$\frac{1}{u} \frac{du}{dx} = y \cdot \frac{1}{\cos x} (-\sin x) + \log \cos x \frac{dy}{dx}$$

$$\begin{aligned}\text{Or, } \frac{du}{dx} &= u \left[-y \frac{\sin x}{\cos x} + \log \cos x \frac{dy}{dx} \right] \\ &= (\cos x)^y \left[-y \tan x + \log \cos x \frac{dy}{dx} \right]\end{aligned}$$

$$\text{Again, } v = (\sin y)^x$$

$$\log v = x \log \sin y$$

$$\frac{1}{v} \frac{dv}{dx} = \left[x \frac{\cos y}{\sin y} \frac{dy}{dx} + \log \sin y \right]$$

$$= (\sin y)^x \left[x \cot y \frac{dy}{dx} + \log \sin y \right]$$

Substituting the value in eqn-1

$$-y \cos x^y \tan x + \cos x^y \log \cos x \frac{dy}{dx} + x (\sin y)^x \cot y \frac{dy}{dx} + (\sin y)^x \log \sin y = 0$$

$$\begin{aligned}\rightarrow \frac{dy}{dx} [\cos x^y \log \cos x + x (\sin y)^x \cot y] &= y \\ (\cos x)^y \tan x - (\sin y)^x \log \sin y\end{aligned}$$

$$\begin{aligned}\rightarrow \frac{dy}{dx} &= \frac{(\cos x)^y y \tan x - (\sin y)^x \log \sin y}{(\cos x)^y \log \cos x + (\sin y)^x x \cot y} \\ \text{(Ans.)}\end{aligned}$$

15. Oct- 17: Find the differentiate coefficient of the following functions with respect to x.

$$y = x^{\ln x} + x^{\cos^{-1} x}$$

Ans. Let, $u = x^{\ln x}$, $v = x^{\cos^{-1} x}$

$$\therefore y = u + v$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \dots \dots \dots (i)$$

Now,

$$u = x^{\ln x}$$

$$\ln u = \ln x^{\ln x} \text{ [taking ln]}$$

$$\ln u = \ln x \cdot \ln x = \ln^2 x$$

Diff w.r. to x,

$$\frac{1}{u} \cdot \frac{du}{dx} = 2 \ln x \cdot \frac{1}{x}$$

$$\frac{du}{dx} = \frac{u}{x} \cdot 2 \ln x$$

$$= \frac{x^{\ln x}}{x} \cdot 2 \ln x$$

Again,

$$v = x^{\cos^{-1} x}$$

$$\ln v = \ln x^{\cos^{-1} x} \text{ [taking ln]}$$

$$\ln v = \cos^{-1} x \cdot \ln x$$

Diff w.r. to x,

$$\frac{1}{v} \cdot \frac{dv}{dx} = \cos^{-1} x \cdot \frac{1}{x} + \ln x \times \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{dv}{dx} = v \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

$$= x^{\cos^{-1} x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

Putting the values of $\frac{du}{dx}$ & $\frac{dv}{dx}$ in ... (i)

$$\frac{dy}{dx} = \frac{x^{\ln x}}{x} \cdot 2 \ln x + x^{\cos^{-1} x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

16. Find $\frac{dy}{dx}$, if $\sin x = \frac{2t}{1+t^2}$ $\tan y = \frac{2t}{1-t^2}$

Given, $\sin x = \frac{2t}{1+t^2}$

Diff w.r. to t,

$$\cos x \frac{dx}{dt} = \frac{(1+t^2) \cdot 2 - 2t \cdot 2t}{(1+t^2)^2}$$

$$\frac{dx}{dt} = \frac{1}{\cos x} \cdot \frac{2-2t^2-4t^2}{(1+t^2)^2}$$

$$= \frac{1}{\sqrt{1-\sin^2 x}} \times \frac{2-2t^2}{(1+t^2)^2}$$

$$= \frac{1}{\sqrt{1-\frac{4t^2}{(1+t^2)^2}}} \times \frac{2(1-t^2)}{(1+t^2)^2}$$

$$= \frac{(1+t^2)}{(1-t^2)} \times \frac{2((1-t^2))}{(1+t^2)^2}$$

$$= \frac{2}{1+t^2}$$

Again, $\tan y = \frac{2t}{1-t^2}$

$$\sec^2 y \frac{dy}{dt} = \frac{(1-t^2) \cdot 2 - 2t \cdot (-2t)}{(1-t^2)^2} = \frac{2-2t^2+4t^2}{(1-t^2)^2}$$

$$= \frac{1}{\sec^2 y} \times \frac{2+2t^2}{(1-t^2)^2}$$

$$= \frac{1}{1+\tan^2 y} \times \frac{2(1+t^2)}{(1-t^2)^2}$$

$$= \frac{1}{1+\frac{4t^2}{(1-t^2)^2}} \times \frac{2(1+t^2)}{(1-t^2)^2}$$

$$= \frac{(1-t^2)^2}{1-2t^2+t^4+4t^2} \times \frac{2(1+t^2)}{(1-t^2)^2}$$

$$= \frac{2(1+t^2)}{(1+t^2)^2}$$

$$= \frac{2}{1+t^2}$$

$$\frac{dy}{dx} = \frac{dt}{dx} \times \frac{dy}{dt}$$

$$= \frac{1+t^2}{2} + \frac{2}{1+t^2}$$

$$= 1 \quad (\text{Ans.})$$

17. If, $x^{\sin x} = y$, $Z = (\sin x)^x$ find $\frac{dy}{dx} = ?$

Given, $x^{\sin x} = y$

$\log y = \sin x \log x$

$$\rightarrow \frac{1}{y} \frac{dy}{dx} = \sin x \cdot \frac{1}{x} + \log x \cos x$$

$$\frac{dy}{dx} = \frac{y}{x} \sin x + y \log x \cos x$$

$$= \frac{x^{\sin x} \sin x}{x} + \log x \cos x \cdot (x^{\sin x})$$

Again, $Z = (\sin x)^x$

$\rightarrow \log Z = x \log \sin x$

$$\rightarrow \frac{1}{Z} \frac{dz}{dx} = x \cdot \frac{1}{\sin x} \cos x + \log \sin x$$

$$\frac{dz}{dx} = Z (x \cot x + \log \sin x)$$

$$= (\sin x)^x (x \cot x + \log \sin x)$$

$$\therefore \frac{dy}{dz} = \frac{dy}{dx} \div \frac{dz}{dx} = \frac{dy}{dx} \times \frac{dx}{dz}$$

$$= \frac{x^{\sin x} (\sin x + x \log x \cos x)}{(\sin x)^x (x^2 \cot x + x \log \sin x)} \text{ (Ans.)}$$

18. Diff $\tan^{-1} \frac{2x}{1-x^2}$ w.r. to $\sin^{-1} \frac{2x}{1+x^2}$

[Nov-21, Oct-18]

Ans:

$$\text{Let, } y = \tan^{-1} \frac{2x}{1-x^2} \text{ [Let } x = \tan \theta]$$

$$\theta = \tan^{-1} x$$

$$= \tan^{-1} \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \tan^{-1} \tan 2\theta$$

$$= 2\theta$$

$$= 2 \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{2}{1+x^2}$$

$$\text{Again, } Z = \sin^{-1} \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$= \sin^{-1} \sin 2\theta$$

$$= 2\theta$$

$$= 2 \tan^{-1} x$$

$$\frac{dz}{dx} = \frac{2}{1+x^2}$$

$$\frac{dy}{dz} = \frac{dy}{dx} \div \frac{dz}{dx} = \frac{dy}{dx} \times \frac{dx}{dz}$$

$$= \frac{1+x^2}{2} \times \frac{2}{1+x^2} = 1 \text{ (Ans.)}$$

19. If, $y = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$, $z = \sec^{-1} \frac{1}{2x^2-1}$ find $\frac{dy}{dz} = ?$

Let, $y = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$, let $x = \sin \theta$, $\theta = \sin^{-1} x$

$$= \tan^{-1} \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \tan^{-1} \tan \theta$$

$$= \theta$$

$$= \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Again,

$$Z = \sec^{-1} \frac{1}{2x^2-1} \text{ [Let, } x = \cos \theta]$$

$$= \sec^{-1} \frac{1}{2\cos^2 \theta - 1}$$

$$= \sec^{-1} \frac{1}{2\cos \theta}$$

$$= 2 \sec^{-1} \sec \theta$$

$$= 2\theta$$

$$= 2 \cos^{-1} x$$

$$\frac{dz}{dx} = \frac{-2}{\sqrt{1-x^2}}$$

$$\therefore \frac{dy}{dz} = \frac{dy}{dx} \div \frac{dz}{dx} = \frac{dy}{dx} \times \frac{dx}{dz}$$

$$= \frac{1}{\sqrt{1-x^2}} \times -\frac{\sqrt{1-x^2}}{2} = -\frac{1}{2} \text{ (Ans)}$$

20. if $x = \frac{3at}{1+t^3}$; $Y = \frac{3at^2}{1+t^2}$ then find $\frac{dy}{dx}$

Given, $x = \frac{3at}{1+t^3}$

$$\frac{dx}{dt} = \frac{(1+t^3) \cdot 3a - 3at \cdot 3t^2}{(1+t^3)^2} = \frac{3a - 6at^3}{(1+t^3)^2} = \frac{3(a - 2t^3)}{(1+t^3)^2}$$

$$\frac{dy}{dt} = \frac{(1+t^2) \cdot 6at - 3at^2 \cdot 2t}{(1+t^2)^2}$$

$$= \frac{6at + 6at^3 - 6at^3}{(1+t^2)^2}$$

$$= \frac{6at}{(1+t^2)^2}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$= \frac{6at}{(1+t^2)^2} \times \frac{(1+t^3)^2}{3a(1-2t^3)} = \frac{2t(1+t^3)^2}{(1+t^2)^2(1-2t^3)} \text{ (Ans)}$$

21 Diff $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ w.r. to x

Ans: Let, $y = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$

Let, $x = \tan \theta$

Then, $Y = \tan^{-1} \frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta}$

$= \tan^{-1} \frac{\sec \theta - 1}{\tan \theta}$

$= \tan^{-1} \frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}}$

$= \tan^{-1} \frac{1 - \cos \theta}{\sin \theta}$

$= \tan^{-1} \frac{2 \sin^2 \theta / 2}{2 \sin \theta / 2 \cos \theta / 2}$

$= \tan^{-1} \tan \theta / 2$

$= \theta / 2 = \frac{\tan^{-1} x}{2}$

$\frac{dy}{dx} = \frac{1}{2} \frac{1}{1+x^2}$ (Ans)

22. if $x = a(\theta + \sin \theta)$, $y = 1 + \cos \theta$

Then Find $\frac{dy}{dx}$ [Oct-17]

$\therefore \frac{dx}{d\theta} = a(1 + \cos \theta)$ $\frac{dy}{d\theta} = -\sin \theta$

Now, $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$

$\frac{dy}{dx} = \frac{-\sin \theta}{a(1 + \cos \theta)}$ (Ans.)

23. Find the Derivative of the parametric equation,

$X = a(t - \sin t)$, $y = a(1 - \cos t)$ [Ap-24]

Ans:

Given, $x = a(t - \sin t)$

$\frac{dx}{dt} = a(1 - \cos t)$

Again $y = a(1 - \cos t)$

$\frac{dy}{dt} = a \sin t$

$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{a \sin t}{a(1 - \cos t)}$

$= \frac{\sin t}{(1 - \cos t)}$ (Ans)

24 Find. The Derivative of the function [Nov-23]

$y = \tan^{-1} \left[\frac{\cos x}{1 + \sin x} \right]$

Ans: $y = \tan^{-1} \left[\frac{\cos x}{1 + \sin x} \right]$

$= \tan^{-1} \left[\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} \right]$

$= \tan^{-1} \left[\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{(\sin \frac{x}{2} + \cos \frac{x}{2})^2} \right]$

$= \tan^{-1} \left[\frac{(\cos \frac{x}{2} + \sin \frac{x}{2})(\cos \frac{x}{2} - \sin \frac{x}{2})}{(\sin \frac{x}{2} + \cos \frac{x}{2})(\sin \frac{x}{2} + \cos \frac{x}{2})} \right]$

$= \tan^{-1} \left[\frac{(\cos \frac{x}{2} - \sin \frac{x}{2})}{(\sin \frac{x}{2} + \cos \frac{x}{2})} \right]$

$= \tan^{-1} \left[\frac{(1 - \tan \frac{x}{2})}{(1 + \tan \frac{x}{2})} \right]$

$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right] \left[\tan \left(\frac{\pi}{4} - \theta \right) = \frac{(1 - \tan \theta)}{(1 + \tan \theta)} \right]$

$Y = \frac{\pi}{4} - \frac{x}{2}$

$\frac{dy}{dx} = -\frac{1}{2}$ (Ans)

25. Find The Derivative of the function

$y = \tan^{-1} \left[\frac{\sin x}{1 + \cos x} \right]$

Ans: $y = \tan^{-1} \left[\frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right]$

$y = \tan^{-1} \left[\frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} \right]$

$= \tan^{-1} \left[\tan \frac{x}{2} \right]$

$Y = x/2$

$\frac{dy}{dx} = \frac{1}{2}$ (Ans)

Partial Differentiation

Q-1. If $v = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$ prove that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = 0$

Solution:

Given that,

$$v = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$$

Diff with w.r.to x

$$\frac{\partial v}{\partial x} = -\frac{z}{x^2} + \frac{1}{y}$$

$$\frac{\partial v}{\partial y} = \frac{1}{z} - \frac{x}{y^2}$$

$$\frac{\partial v}{\partial z} = -\frac{y}{z^2} + \frac{1}{x}$$

we get,

$$\begin{aligned} x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} &= -\frac{z}{x} + \frac{x}{y} + \frac{y}{z} - \frac{x}{y} - \frac{y}{z} + \frac{z}{x} \\ &= 0 \text{ (Proved)} \end{aligned}$$

Q-2. Show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$ If (i) $v = \log(x^2 + y^2)$ (ii) $v = \tan^{-1} \left(\frac{y}{x} \right)$

Solution: Given that, $v = \log(x^2 + y^2)$

Diff w.r. to x,

$$\frac{\partial v}{\partial x} = \frac{2x}{x^2 + y^2}$$

$$\therefore \frac{\partial^2 v}{\partial x^2} = \frac{(x^2 + y^2) \cdot 2 - 2x \cdot 2x}{(x^2 + y^2)^2} = \frac{2x^2 + 2y^2 - 4x^2}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2} \dots (i)$$

Diff w.r.to y,

$$\frac{\partial v}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\therefore \frac{\partial^2 v}{\partial y^2} = \frac{(x^2 + y^2) \cdot 2 - 2y \cdot 2y}{(x^2 + y^2)^2} = \frac{2x^2 + 2y^2 - 4y^2}{(x^2 + y^2)^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2} \dots (ii)$$

Adding (i) + (ii)

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2} + \frac{2x^2 - 2y^2}{(x^2 + y^2)^2} = 0 \quad (\text{proved})$$

(ii) $v = \tan^{-1} \frac{y}{x}$

diff w.r. to x,

$$\begin{aligned} \frac{\partial v}{\partial x} &= \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot -\frac{y}{x^2} \\ &= \frac{x^2}{x^2 + y^2} \cdot -\frac{y}{x^2} \\ &= \frac{-y}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \left\{ \frac{(x^2 + y^2) \cdot 0 - (-y) \frac{\partial}{\partial x} (x^2 + y^2)}{(x^2 + y^2)^2} \right\} \\ &= \frac{2xy}{(x^2 + y^2)^2} \quad \dots \dots (1) \end{aligned}$$

diff w. r. to y $\frac{\partial v}{\partial y} = \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x}$

$$\begin{aligned} \frac{\partial v}{\partial y} &= \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} \\ &= \frac{x}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 v}{\partial y^2} &= \frac{(x^2 + y^2) \cdot 0 - x \cdot 2y}{(x^2 + y^2)^2} \\ &= \frac{-2xy}{(x^2 + y^2)^2} \quad \dots \dots \dots (2) \end{aligned}$$

By adding (1) & (2) we get,

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{2xy}{(x^2 + y^2)^2} - \frac{2xy}{(x^2 + y^2)^2}$$

Q-3. If $v = z \tan^{-1} \left(\frac{y}{x} \right)$, then prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

Solution: Given that,

$$\begin{aligned} \frac{\partial v}{\partial x} &= z \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot -\frac{y}{x^2} \\ &= -\frac{x^2 z}{x^2 + y^2} \cdot \frac{y}{x^2} \\ &= -\frac{yz}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= -\left\{ \frac{(x^2 + y^2) \cdot 0 - yz \cdot 2x}{(x^2 + y^2)^2} \right\} \\ &= \frac{2xyz}{(x^2 + y^2)^2} \quad \dots \dots (i) \end{aligned}$$